Review: Identifying congruent triangles

1 **A** and **B**

2 **B** and **D**

3
- a
  - i **B** and **C**
  - ii SSS: All three corresponding sides are congruent.
- b
  - i **A** and **C**
  - ii SAS: A pair of corresponding sides and the included angle are congruent.
- c
  - i **B** and **C**
  - ii ASA: A pair of corresponding angles and the included side are congruent.
- d
  - i **A** and **C**
  - ii SAS: A pair of corresponding sides and the included angle are congruent.

4
- a
  - i **A** and **B**
  - ii SAS: A pair of corresponding sides and the included angle are congruent.
- b
  - i **B** and **C**
  - ii AAS: A pair of corresponding angles and a non-included side are congruent.
- c
  - i **C** and **D**
  - ii HL: Two right-angled triangles with hypotenuse and one leg are congruent.
- d
  - i **C** and **D**
  - ii SAS: A pair of corresponding sides and the included angle are congruent.

5
- a No
- b No
- c No
- d No

6 No

7 No

8

a $\overline{DF}$

b SSS: All three corresponding sides are congruent.

9

a $\angle CDB$

b SAS: A pair of corresponding sides and the included angle are congruent.

10

a $\angle RQS$

b ASA: A pair of corresponding angles and the included side are congruent.

11

a $\overline{PN}$

b SSS: All three corresponding sides are congruent.

12

a $\angle CDB$

b SAS: A pair of corresponding sides and the included angle are congruent.

13

a $\angle POQ$

b AAS: A pair of corresponding angles and a non-included side are congruent.

14

a $\triangle PRN$ and $\triangle QRM$

b $\triangle EAB$ and $\triangle BDE$

15 One triangle could be an enlarged or reduced version of the other.

16 No, not always.

Additional questions

17

a. $\triangle ABC$ and $\triangle EDC$ are congruent. $\triangle EDC$ is a rotation of $\triangle ABC$ by 180° around point C.

b. $\triangle PQR$ and $\triangle SQR$ are congruent. $\triangle SQR$ is a reflection of $\triangle PQR$ across $\overleftrightarrow{QR}$.

c. $\triangle UVW$ and $\triangle XYZ$ are not congruent.

18

a. Congruent by Side-angle-side congruence (SAS)

b. Congruent by Side-side-side congruence (SSS)

c. Not congruent

d. Not enough information

e. Congruent by Angle-angle-side congruence (AAS)

f. Not congruent

g. Not enough information

h. Not congruent

i. $\triangle ABC \cong \triangle DEF$ based on the HL congruence theorem.

j. There is not enough information to determine whether $\triangle GHJ \cong \triangle KLM$ are congruent.

k. $\triangle ABC$ and $\triangle STU$ are not congruent.

l. $\triangle PQR$ and $\triangle STU$ are congruent based on the Side-angle-side (SAS) congruence theorem.

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a. $x = 11.5$

b. $x = 90^{\circ}$

c. $x = 7$

d. $x = 3$

e. $x = 9$

f. $x = 4$

g. $x = 26$

h. $x = 60^{\circ}$

i. $x = 19$ and $y = 18$

20

a. There is enough information for Side-angle-side (SAS) congruence

b. Answers may vary.
There is not enough information for congruence. If $\angle E \cong \angle B$ the triangles would be congruent by Side-angle-side congruence (SAS).

c There is not enough information for con[illegible]ce. If $\overline{EG} \cong \overline{HK}$ the triangles would be congruent by Side-side-side congruence (SSS) or if $\angle F \cong \angle J$ the triangles would be congruent by Side-angle-side congruence (SAS).

d There is enough information for Side-side-side congruence (SSS).

21

a
- i $\triangle ACD$ and $\triangle BCE$
- ii Side-side-side congruence (SSS)

b Answers may vary:
- i $\triangle PRN$ and $\triangle QRM$
- ii Side-side-side congruence (SSS) or Side-angle-side congruence (SAS)

c Answers may vary:
- i $\triangle VZY \cong \triangle YXV$
- ii Hypotenuse-leg (HL)

22

a
- i $\angle C \cong \angle B$
- ii $\angle BAD \cong \angle CAD$

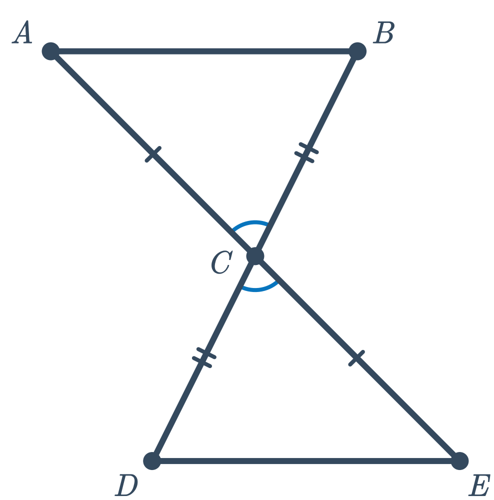
b
- i Either $\overline{AC} \cong \overline{JM}$ or $\overline{AB} \cong \overline{KM}$
- ii $\overline{BC} \cong \overline{JK}$

23

a $\overline{UV} \cong \overline{WX}$

b $\angle CED \cong \angle FDE$

24 Yes. The intersecting lines form a pair of congruent vertical angles. Because the segments bisect each other, the triangles will be congruent by Side-angle-side congruence (SAS) no matter how long the segments are.



25 No. While it is true that the longest side of a [illegible]-triangle is the hypotenuse, the triangles that Fleur and Jeung have drawn are not necessarily right-triangles. Just knowing that two pairs of sides are congruent is not enough to prove congruency.

26 Yes. If the side is in between the two given angles students will be able to construct a congruent triangle by Angle-side-angle congruence (ASA) and if the side is *not* between the two angles the triangles will be congruent by Angle-angle-side congruence (AAS).